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React JS Lesson 17 – React Routing (Premium Notes)

1) What is Routing in React

- Routing means moving between pages without full page reload.
- React uses React Router library for routing.

Example:

Home → About → Contact

Explanation:

- Page changes happen inside same app.

2) Single Page Application (SPA) Concept

- SPA loads one HTML page once.
- Only content changes during navigation.

Example:

Browser loads once → content changes dynamically

Explanation:

- Fast navigation experience.

3) Installing React Router

- React Router must be installed before use.

Example:

```
npm install react-router-dom
```

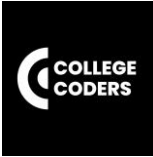
Explanation:

- Package adds routing support.

4) BrowserRouter Setup

- BrowserRouter wraps full app.

Example:



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```
import { BrowserRouter } from 'react-router-dom';
```

```
<BrowserRouter>  
  <App />  
</BrowserRouter>
```

Explanation:

- Routing becomes active globally.

5) Routes and Route

- Routes contains all route definitions.
- Route connects path with component.

Example:

```
<Routes>  
  <Route path="/" element={<Home />} />  
</Routes>
```

Explanation:

- Home renders on / path.

6) Creating Multiple Pages

- Each page is separate component.

Example:

```
<Route path="/about" element={<About />} />  
<Route path="/contact" element={<Contact />} />
```

Explanation:

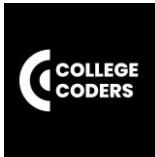
- Each URL loads matching component.

7) Navigation using Link / NavLink

- Link changes page without reload.
- NavLink helps active link styling.

Example:

```
<Link to="/about">About</Link>
```



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Explanation:

- No browser refresh happens.

8) 404 Page (Not Found)

- Unknown route should show custom page.

Example:

```
<Route path='*' element={<NotFound />} />
```

Explanation:

- * catches unmatched routes.

Mini Project – 3 Page Navigation

Home.jsx

About.jsx

Contact.jsx

Navbar with Link components

- Create three pages.
- Add navbar links.
- Add 404 page.